

Helios Compute Control - technical appendix

Illustrative synthetic case - not investment advice

This appendix retains reproducibility records, raw identifiers, mappings, and formula definitions. It is not the investment-committee front page.

Content identity

- Manifest SHA-256: 4f590ab6ea0b33914062348b7e3bdcca71475bb97a2e4c85b9594a94b4fe17f4
- Analysis SHA-256: 39bd296eddbdf32b39c3070d744824a84d2469fa6028aff823670ea7e19a71072
- Decision SHA-256: 501f4a36e3b48b5fa96c5bec789d3643a3313240216c2a120114049112fb22cd
- Packet SHA-256: 392b74abebaf1b51fa5b3913b382fb67e6e4058d04e598bbca893b78f8a8841

Analysis receipts

ANALYSIS	CLASSIFICATION	STATE	SPECIFICATION	RECEIPT
HX-01	ACCOUNTING_IDENTITY	REPORTED	8c611e82c119a76c9c45f8244d4960b22058ffc883c32b3d93482017cb5855c0	b9c07b32c1a6262f3b0069a79be82d445a7701b8ad77f7d094ff773b90251a32
HX-02	DESCRIPTIVE	REPORTED	4dd16a8f2fab9ac0217da45fddb3156f53c234ee6c3b1e1143bala4338101858	0509303169f8bf025bc3594b4139e2f62a7cf061e639ee3eef933039118fd724
HX-03	ACCOUNTING_IDENTITY	REPORTED	3527bf827d0ab45398cc7c9811fd701083e2a5be358113alf366f2fcc6fc332	5480f1ff78f5c6c51b38a0fce36fd36204df4e491d71e8e2d779c1b342949c5c
HX-04	DESCRIPTIVE	REPORTED	e388f757bbcddee972471756401e50cabb2389cb7b5ef90f2e555eb7dceb02a0a	0fe11718ecad1d1e6415d00cfcb3efb9eab35f2931ae10elda69ba79cb5e326a
HX-05	PREDICTIVE_ASSOCIATION	REPORTED	869fe38c82c8037542206b8d3c5a0646792563b18f4f414a75cebe01996becb4	56ae7b892ca96d92b28b10e2894be50d26850174166ef3d4a4210a432a469216
HX-06	CAUSAL_SYNTHETIC_ONLY	REPORTED	0e1b65ba0e8dee3c2a13481cb97850a5a43ce3ab6ac091b2678138c2010e6f2b	13ce7025a4507d27baea7edc84ff4f3dde9478bafe4ce720e87899fa8a8f6c20
HX-07	NOT_IDENTIFIED	ABSTAIN	1d274869a7fe75ea3c635061cb3dd615d1ceb7a273b83d4917d477f7f6515e3cc	73fa9bef8097c9ee38a1100648e9c5f18cd6e8856ad5aedd9ba839e3836809f7
HX-08	ACCOUNTING_IDENTITY	REPORTED	a837153dd36285f132e0c74662d034ab3543b5be1a4d011d069d78086fe9ca29	9fff1fd75cee1f45e1913ad9dc9cf96f816b87ee795247085f49c6ef6957067f
HX-09	SCENARIO	REPORTED	7780c967d835a7e2f5e001adc48968e348de043ce81d2ec84707a7f1ca315bfe	814cbc3df911d6fa721a2e4232410f7e429d433def1aace7312494c88753d634

Evidence-to-model mappings

ANALYSIS	CREDIT CLASS	OBSERVED VALUE	MODEL TREATMENT
HX-05	SCENARIO_ONLY	\$20.90B modeled serviceable spend	Scenario calibration only; tier five remains unidentified
HX-06	VALUE_CREATION_BRIDGE	-0.0911 log-point synthetic ITT	No base-case credit; adoption and valuation remain scenario judgments
HX-01	BASE_CASE	ltm_revenue: 5695078389 cents	Bounded base-case evidence
HX-02	BASE_CASE	pooled_nrr: 135.36 percent	Bounded base-case evidence
HX-03	BASE_CASE	burn_multiple: 0.78 multiple	Bounded base-case evidence
HX-04	ZERO	inflated_opportunities: 48 count	0
HX-07	ZERO	No estimate retained	0
HX-08	BASE_CASE	first_close_new_shares: 1814223 shares	Bounded base-case evidence
HX-09	SCENARIO_ONLY	p10_moic: 0.00 multiple	Scenario calibration only

Formula register

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-headline-ownership	DIVIDE	hx-ownership	helios-MILESTONE-target-shares, helios-MILESTONE-fully-diluted-shares
vc-formula-base-bridge-exit-ev	MULTIPLY	helios-BASE-bridge-exit-enterprise-value	helios-BASE-bridge-terminal-revenue, helios-BASE-bridge-exit-multiple
vc-formula-base-bridge-exit-equity	ADD	helios-BASE-bridge-exit-equity	helios-BASE-bridge-exit-enterprise-value, helios-BASE-bridge-exit-cash
vc-formula-base-bridge-exit-cash	ADD	helios-BASE-bridge-exit-cash	helios-BASE-month-60-ending_cash_cents, helios-BASE-bridge-exit-cash-zero
vc-formula-base-minimum-cash	MIN	helios-BASE-minimum-cash	helios-BASE-month-01-ending_cash_cents, helios-BASE-month-02-ending_cash_cents, helios-BASE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 10c65ce3fbd8c2c6717cbb53a28d45bae7c11395c4a7a5a98c1642c608c1b36)
vc-formula-base-funded-capital	SUM	helios-BASE-target-invested	helios-BASE-event-series-c-close-new-money, helios-BASE-event-series-c-tranche-new-money
vc-formula-base-ownership	DIVIDE	helios-BASE-ownership	helios-BASE-target-shares, helios-BASE-fully-diluted-shares
vc-formula-base-moic	DIVIDE	helios-BASE-gross-moic	helios-BASE-target-proceeds, helios-BASE-target-invested
vc-formula-base-dated-xirr	DATED_XIRR	helios-BASE-gross-xirr	helios-BASE-target-cash-flow-01, helios-BASE-target-cash-flow-02
vc-formula-base-target-proceeds	ADD	helios-BASE-target-proceeds	helios-BASE-waterfall-series_c-proceeds, helios-BASE-target-proceeds-zero
vc-formula-base-waterfall-conservation	SUM	helios-BASE-exit-value	helios-BASE-waterfall-common, helios-BASE-waterfall-series_a-proceeds, helios-BASE-waterfall-series_b-proceeds, helios-BASE-waterfall-series_c-proceeds, helios-BASE-waterfall-series_d-proceeds
vc-formula-milestone-bridge-exit-ev	MULTIPLY	helios-MILESTONE-bridge-exit-enterprise-value	helios-MILESTONE-bridge-terminal-revenue, helios-MILESTONE-bridge-exit-multiple
vc-formula-milestone-bridge-exit-equity	ADD	helios-MILESTONE-bridge-exit-equity	helios-MILESTONE-bridge-exit-enterprise-value, helios-MILESTONE-bridge-exit-cash
vc-formula-milestone-bridge-exit-cash	ADD	helios-MILESTONE-bridge-exit-cash	helios-MILESTONE-month-60-ending_cash_cents, helios-MILESTONE-bridge-exit-cash-zero
vc-formula-milestone-minimum-cash	MIN	helios-MILESTONE-minimum-cash	helios-MILESTONE-month-01-ending_cash_cents, helios-MILESTONE-month-02-ending_cash_cents, helios-MILESTONE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 a99f723cf3844e3b311a749dc4283338a491d6d3a3cdde22b48892da3112ce17)
vc-formula-milestone-funded-capital	SUM	helios-MILESTONE-target-invested	helios-MILESTONE-event-series-c-close-new-money, helios-MILESTONE-event-series-c-tranche-new-money
vc-formula-milestone-ownership	DIVIDE	helios-MILESTONE-ownership	helios-MILESTONE-target-shares, helios-MILESTONE-fully-diluted-shares
vc-formula-milestone-moic	DIVIDE	helios-MILESTONE-gross-moic	helios-MILESTONE-target-proceeds, helios-MILESTONE-target-invested
vc-formula-milestone-dated-xirr	DATED_XIRR	helios-MILESTONE-gross-xirr	helios-MILESTONE-target-cash-flow-01, helios-MILESTONE-target-cash-flow-02, helios-MILESTONE-target-cash-flow-03
vc-formula-milestone-01-funding	ADD	helios-MILESTONE-month-01-cash-before-operations	helios-MILESTONE-month-01-beginning_cash_cents, helios-MILESTONE-month-01-financing_cash_cents
vc-formula-milestone-01-ending	ADD	helios-MILESTONE-month-01-ending-cash-recomputed	helios-MILESTONE-month-01-cash-before-operations, helios-MILESTONE-month-01-operating_net_cash_flow_cents
vc-formula-milestone-12-funding	ADD	helios-MILESTONE-month-12-cash-before-operations	helios-MILESTONE-month-12-beginning_cash_cents, helios-MILESTONE-month-12-financing_cash_cents
vc-formula-milestone-12-ending	ADD	helios-MILESTONE-month-12-ending-cash-recomputed	helios-MILESTONE-month-12-cash-before-operations, helios-MILESTONE-month-12-operating_net_cash_flow_cents
vc-formula-milestone-24-funding	ADD	helios-MILESTONE-month-24-cash-before-operations	helios-MILESTONE-month-24-beginning_cash_cents, helios-MILESTONE-month-24-financing_cash_cents
vc-formula-milestone-24-ending	ADD	helios-MILESTONE-month-24-ending-cash-recomputed	helios-MILESTONE-month-24-cash-before-operations, helios-MILESTONE-month-24-operating_net_cash_flow_cents
vc-formula-milestone-36-funding	ADD	helios-MILESTONE-month-36-cash-before-operations	helios-MILESTONE-month-36-beginning_cash_cents, helios-MILESTONE-month-36-financing_cash_cents
vc-formula-milestone-36-ending	ADD	helios-MILESTONE-month-36-ending-cash-recomputed	helios-MILESTONE-month-36-cash-before-operations, helios-MILESTONE-month-36-operating_net_cash_flow_cents
vc-formula-milestone-60-funding	ADD	helios-MILESTONE-month-60-cash-before-operations	helios-MILESTONE-month-60-beginning_cash_cents, helios-MILESTONE-month-60-financing_cash_cents
vc-formula-milestone-60-ending	ADD	helios-MILESTONE-month-60-ending-cash-recomputed	helios-MILESTONE-month-60-cash-before-operations, helios-MILESTONE-month-60-operating_net_cash_flow_cents

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-milestone-target-proceeds	ADD	helios-MILESTONE-target-proceeds	helios-MILESTONE-waterfall-series_c-proceeds, helios-MILESTONE-target-proceeds-zero
vc-formula-milestone-waterfall-conservation	SUM	helios-MILESTONE-exit-value	helios-MILESTONE-waterfall-common, helios-MILESTONE-waterfall-series_a-proceeds, helios-MILESTONE-waterfall-series_b-proceeds, helios-MILESTONE-waterfall-series_c-proceeds
vc-formula-downside-bridge-exit-ev	MULTIPLY	helios-DOWNSIDE-bridge-exit-enterprise-value	helios-DOWNSIDE-bridge-terminal-revenue, helios-DOWNSIDE-bridge-exit-multiple
vc-formula-downside-bridge-exit-equity	ADD	helios-DOWNSIDE-bridge-exit-equity	helios-DOWNSIDE-bridge-exit-enterprise-value, helios-DOWNSIDE-bridge-exit-cash
vc-formula-downside-bridge-exit-cash	ADD	helios-DOWNSIDE-bridge-exit-cash	helios-DOWNSIDE-month-60-ending_cash_cents, helios-DOWNSIDE-bridge-exit-cash-zero
vc-formula-downside-minimum-cash	MIN	helios-DOWNSIDE-minimum-cash	helios-DOWNSIDE-month-01-ending_cash_cents, helios-DOWNSIDE-month-02-ending_cash_cents, helios-DOWNSIDE-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 d4d09f9e067397a980660222cb6b8c58797ddd65183199349174708e1828e70f)
vc-formula-downside-funded-capital	SUM	helios-DOWNSIDE-target-invested	helios-DOWNSIDE-event-series-c-close-new-money, helios-DOWNSIDE-event-series-c-tranche-new-money
vc-formula-downside-ownership	DIVIDE	helios-DOWNSIDE-ownership	helios-DOWNSIDE-target-shares, helios-DOWNSIDE-fully-diluted-shares
vc-formula-downside-moic	DIVIDE	helios-DOWNSIDE-gross-moic	helios-DOWNSIDE-target-proceeds, helios-DOWNSIDE-target-invested
vc-formula-downside-dated-xirr	DATED_XIRR	helios-DOWNSIDE-gross-xirr	helios-DOWNSIDE-target-cash-flow-01, helios-DOWNSIDE-target-cash-flow-02
vc-formula-downside-target-proceeds	ADD	helios-DOWNSIDE-target-proceeds	helios-DOWNSIDE-waterfall-series_c-proceeds, helios-DOWNSIDE-target-proceeds-zero
vc-formula-downside-waterfall-conservation	SUM	helios-DOWNSIDE-exit-value	helios-DOWNSIDE-waterfall-common, helios-DOWNSIDE-waterfall-series_a-proceeds, helios-DOWNSIDE-waterfall-series_b-proceeds, helios-DOWNSIDE-waterfall-series_c-proceeds, helios-DOWNSIDE-waterfall-series_d-proceeds
vc-formula-financing_shortfall-bridge-exit-ev	MULTIPLY	helios-FINANCING_SHORTFALL-bridge-exit-enterprise-value	helios-FINANCING_SHORTFALL-bridge-terminal-revenue, helios-FINANCING_SHORTFALL-bridge-exit-multiple
vc-formula-financing_shortfall-bridge-exit-equity	ADD	helios-FINANCING_SHORTFALL-bridge-exit-equity	helios-FINANCING_SHORTFALL-bridge-exit-enterprise-value, helios-FINANCING_SHORTFALL-bridge-exit-cash
vc-formula-financing_shortfall-bridge-exit-cash	ADD	helios-FINANCING_SHORTFALL-bridge-exit-cash	helios-FINANCING_SHORTFALL-month-60-ending_cash_cents, helios-FINANCING_SHORTFALL-bridge-exit-cash-zero
vc-formula-financing_shortfall-minimum-cash	MIN	helios-FINANCING_SHORTFALL-minimum-cash	helios-FINANCING_SHORTFALL-month-01-ending_cash_cents, helios-FINANCING_SHORTFALL-month-02-ending_cash_cents, helios-FINANCING_SHORTFALL-month-03-ending_cash_cents; +57 more in model-appendix.json (ordered-list SHA-256 e633a4835ald7e7b9bf93a0f30d81076e6b835c529a087137e07b29b16f6db4)
vc-formula-financing_shortfall-funded-capital	SUM	helios-FINANCING_SHORTFALL-target-invested	helios-FINANCING_SHORTFALL-event-series-c-close-new-money, helios-FINANCING_SHORTFALL-event-series-c-tranche-new-money
vc-formula-financing_shortfall-ownership	DIVIDE	helios-FINANCING_SHORTFALL-ownership	helios-FINANCING_SHORTFALL-target-shares, helios-FINANCING_SHORTFALL-fully-diluted-shares
vc-formula-financing_shortfall-moic	DIVIDE	helios-FINANCING_SHORTFALL-gross-moic	helios-FINANCING_SHORTFALL-target-proceeds, helios-FINANCING_SHORTFALL-target-invested
vc-formula-financing_shortfall-dated-xirr	DATED_XIRR	helios-FINANCING_SHORTFALL-gross-xirr	helios-FINANCING_SHORTFALL-target-cash-flow-01, helios-FINANCING_SHORTFALL-target-cash-flow-02
vc-formula-financing_shortfall-target-proceeds	ADD	helios-FINANCING_SHORTFALL-target-proceeds	helios-FINANCING_SHORTFALL-waterfall-series_c-proceeds, helios-FINANCING_SHORTFALL-target-proceeds-zero
vc-formula-financing_shortfall-waterfall-conservation	SUM	helios-FINANCING_SHORTFALL-exit-value	helios-FINANCING_SHORTFALL-waterfall-common, helios-FINANCING_SHORTFALL-waterfall-series_a-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_b-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_bridge-proceeds, helios-FINANCING_SHORTFALL-waterfall-series_c-proceeds
vc-formula-distribution-p10-moic	QUANTILE_P10	helios-distribution-moic-0	helios-distribution-draw-count, helios-distribution-p10-rank-index, helios-distribution-path-000-moic; +999 more in model-appendix.json (ordered-list SHA-256 39805f22ed2f5e866b7db01bb1beaab83a0731e45be30f68dec52ba8a78117a)
vc-formula-distribution-p50-moic	QUANTILE_P50	helios-distribution-moic-1	helios-distribution-draw-count, helios-distribution-p50-rank-index, helios-distribution-path-000-moic; +999 more in model-appendix.json (ordered-list SHA-256 6ad63cf6affc72fc1b8004bec20070aa27be997455204ba004994204919dcdcb)
vc-formula-distribution-p90-moic	QUANTILE_P90	helios-distribution-moic-2	helios-distribution-draw-count, helios-distribution-p90-rank-index, helios-distribution-path-000-moic; +999 more in model-appendix.json (ordered-list SHA-256 8d79186813a9d4747f00b1e5943ea091ac1e21b5c08ef95ac1e9b898c3d9465c)
vc-formula-distribution-p10-xirr	QUANTILE_P10	helios-distribution-xirr-0	helios-distribution-draw-count, helios-distribution-p10-rank-index, helios-distribution-path-000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 da8d62f556120bff66818a7e395ae71fb16d4992fad18578de078a7b85d4bb8)

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-distribution-p50-xirr	QUANTILE_P50	helios-distribution-xirr-1	helios-distribution-draw-count, helios-distribution-p50-rank-index, helios-distribution-path-0000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 87161fe1577f84a558f170711ad1fa86c7939c6388e24c0ffe6f688d48b853aa)
vc-formula-distribution-p90-xirr	QUANTILE_P90	helios-distribution-xirr-2	helios-distribution-draw-count, helios-distribution-p90-rank-index, helios-distribution-path-0000-xirr; +999 more in model-appendix.json (ordered-list SHA-256 5c214db2af3223f647e8332e3687d52ebe7cf31f176770f24e72783108d21660)
vc-formula-distribution-probability-below-one	PROBABILITY_BELOW_ONE_PERCENT	helios-hx-09-probability_below_1x	helios-distribution-path-0000-moic, helios-distribution-path-0001-moic, helios-distribution-path-0002-moic; +997 more in model-appendix.json (ordered-list SHA-256 464817cdcae81f9d141d174fd02d354b7bd3d90d3e00970ffe1e5b6d3186626)
vc-formula-vc-annual_revenue_growth-1-moic	DIVIDE	helios-vc-annual_revenue_growth-1-gross-moic	helios-vc-annual_revenue_growth-1-proceeds, helios-vc-annual_revenue_growth-1-invested
vc-formula-vc-annual_revenue_growth-1-xirr	DATED_XIRR	helios-vc-annual_revenue_growth-1-gross-xirr	helios-vc-annual_revenue_growth-1-cash-flow-01, helios-vc-annual_revenue_growth-1-cash-flow-02, helios-vc-annual_revenue_growth-1-cash-flow-03
vc-formula-vc-annual_revenue_growth-1-ownership	DIVIDE	helios-vc-annual_revenue_growth-1-ownership	helios-vc-annual_revenue_growth-1-target-shares, helios-vc-annual_revenue_growth-1-fully-diluted-shares
vc-formula-vc-annual_revenue_growth-1-minimum-cash	MIN	helios-vc-annual_revenue_growth-1-minimum-cash	helios-vc-annual_revenue_growth-1-ending-cash-01, helios-vc-annual_revenue_growth-1-ending-cash-02, helios-vc-annual_revenue_growth-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 62d29f2ba84b8dacadcealfd1db06ed0b9f09b67d6321d20972a7833d644b69)
vc-formula-vc-annual_revenue_growth-1-exit-ev	MULTIPLY	helios-vc-annual_revenue_growth-1-exit-enterprise-value	helios-vc-annual_revenue_growth-1-terminal-revenue, helios-vc-annual_revenue_growth-1-exit-revenue-multiple
vc-formula-vc-annual_revenue_growth-1-exit-equity	ADD	helios-vc-annual_revenue_growth-1-exit-equity-value	helios-vc-annual_revenue_growth-1-exit-enterprise-value, helios-vc-annual_revenue_growth-1-ending-cash-60
vc-formula-vc-annual_revenue_growth-2-moic	DIVIDE	helios-vc-annual_revenue_growth-2-gross-moic	helios-vc-annual_revenue_growth-2-proceeds, helios-vc-annual_revenue_growth-2-invested
vc-formula-vc-annual_revenue_growth-2-xirr	DATED_XIRR	helios-vc-annual_revenue_growth-2-gross-xirr	helios-vc-annual_revenue_growth-2-cash-flow-01, helios-vc-annual_revenue_growth-2-cash-flow-02, helios-vc-annual_revenue_growth-2-cash-flow-03
vc-formula-vc-annual_revenue_growth-2-ownership	DIVIDE	helios-vc-annual_revenue_growth-2-ownership	helios-vc-annual_revenue_growth-2-target-shares, helios-vc-annual_revenue_growth-2-fully-diluted-shares
vc-formula-vc-annual_revenue_growth-2-minimum-cash	MIN	helios-vc-annual_revenue_growth-2-minimum-cash	helios-vc-annual_revenue_growth-2-ending-cash-01, helios-vc-annual_revenue_growth-2-ending-cash-02, helios-vc-annual_revenue_growth-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 ca58f9955c91486a705c4b0d9550758163d05137147f07c4714188425ae771fb)
vc-formula-vc-annual_revenue_growth-2-exit-ev	MULTIPLY	helios-vc-annual_revenue_growth-2-exit-enterprise-value	helios-vc-annual_revenue_growth-2-terminal-revenue, helios-vc-annual_revenue_growth-2-exit-revenue-multiple
vc-formula-vc-annual_revenue_growth-2-exit-equity	ADD	helios-vc-annual_revenue_growth-2-exit-equity-value	helios-vc-annual_revenue_growth-2-exit-enterprise-value, helios-vc-annual_revenue_growth-2-ending-cash-60
vc-formula-vc-annual_revenue_growth-3-moic	DIVIDE	helios-vc-annual_revenue_growth-3-gross-moic	helios-vc-annual_revenue_growth-3-proceeds, helios-vc-annual_revenue_growth-3-invested
vc-formula-vc-annual_revenue_growth-3-xirr	DATED_XIRR	helios-vc-annual_revenue_growth-3-gross-xirr	helios-vc-annual_revenue_growth-3-cash-flow-01, helios-vc-annual_revenue_growth-3-cash-flow-02, helios-vc-annual_revenue_growth-3-cash-flow-03
vc-formula-vc-annual_revenue_growth-3-ownership	DIVIDE	helios-vc-annual_revenue_growth-3-ownership	helios-vc-annual_revenue_growth-3-target-shares, helios-vc-annual_revenue_growth-3-fully-diluted-shares
vc-formula-vc-annual_revenue_growth-3-minimum-cash	MIN	helios-vc-annual_revenue_growth-3-minimum-cash	helios-vc-annual_revenue_growth-3-ending-cash-01, helios-vc-annual_revenue_growth-3-ending-cash-02, helios-vc-annual_revenue_growth-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 b375d6f1a7d8ffa37743465880e385a5b736c97e9c24cd69285c144a2aacdce)
vc-formula-vc-annual_revenue_growth-3-exit-ev	MULTIPLY	helios-vc-annual_revenue_growth-3-exit-enterprise-value	helios-vc-annual_revenue_growth-3-terminal-revenue, helios-vc-annual_revenue_growth-3-exit-revenue-multiple
vc-formula-vc-annual_revenue_growth-3-exit-equity	ADD	helios-vc-annual_revenue_growth-3-exit-equity-value	helios-vc-annual_revenue_growth-3-exit-enterprise-value, helios-vc-annual_revenue_growth-3-ending-cash-60
vc-formula-vc-exit_revenue_multiple-1-moic	DIVIDE	helios-vc-exit_revenue_multiple-1-gross-moic	helios-vc-exit_revenue_multiple-1-proceeds, helios-vc-exit_revenue_multiple-1-invested
vc-formula-vc-exit_revenue_multiple-1-xirr	DATED_XIRR	helios-vc-exit_revenue_multiple-1-gross-xirr	helios-vc-exit_revenue_multiple-1-cash-flow-01, helios-vc-exit_revenue_multiple-1-cash-flow-02, helios-vc-exit_revenue_multiple-1-cash-flow-03
vc-formula-vc-exit_revenue_multiple-1-ownership	DIVIDE	helios-vc-exit_revenue_multiple-1-ownership	helios-vc-exit_revenue_multiple-1-target-shares, helios-vc-exit_revenue_multiple-1-fully-diluted-shares

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-vc-exit_revenue_multiple-1-minimum-cash	MIN	helios-vc-exit_revenue_multiple-1-minimum-cash	helios-vc-exit_revenue_multiple-1-ending-cash-01, helios-vc-exit_revenue_multiple-1-ending-cash-02, helios-vc-exit_revenue_multiple-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 379b2584e8b432470c3b27c5f575a106cabc9cf40c9a2d4859c9f748e15a2b9e)
vc-formula-vc-exit_revenue_multiple-1-exit-ev	MULTIPLY	helios-vc-exit_revenue_multiple-1-exit-enterprise-value	helios-vc-exit_revenue_multiple-1-terminal-revenue, helios-vc-exit_revenue_multiple-1-exit-revenue-multiple
vc-formula-vc-exit_revenue_multiple-1-exit-equity	ADD	helios-vc-exit_revenue_multiple-1-exit-equity-value	helios-vc-exit_revenue_multiple-1-exit-enterprise-value, helios-vc-exit_revenue_multiple-1-ending-cash-60
vc-formula-vc-exit_revenue_multiple-2-moic	DIVIDE	helios-vc-exit_revenue_multiple-2-gross-moic	helios-vc-exit_revenue_multiple-2-proceeds, helios-vc-exit_revenue_multiple-2-invested
vc-formula-vc-exit_revenue_multiple-2-xirr	DATED_XIRR	helios-vc-exit_revenue_multiple-2-gross-xirr	helios-vc-exit_revenue_multiple-2-cash-flow-01, helios-vc-exit_revenue_multiple-2-cash-flow-02, helios-vc-exit_revenue_multiple-2-cash-flow-03
vc-formula-vc-exit_revenue_multiple-2-ownership	DIVIDE	helios-vc-exit_revenue_multiple-2-ownership	helios-vc-exit_revenue_multiple-2-target-shares, helios-vc-exit_revenue_multiple-2-fully-diluted-shares
vc-formula-vc-exit_revenue_multiple-2-minimum-cash	MIN	helios-vc-exit_revenue_multiple-2-minimum-cash	helios-vc-exit_revenue_multiple-2-ending-cash-01, helios-vc-exit_revenue_multiple-2-ending-cash-02, helios-vc-exit_revenue_multiple-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 277fb7d944fd393f48fb91f7158ae933b82d042f934edf61692b99c2ca51109)
vc-formula-vc-exit_revenue_multiple-2-exit-ev	MULTIPLY	helios-vc-exit_revenue_multiple-2-exit-enterprise-value	helios-vc-exit_revenue_multiple-2-terminal-revenue, helios-vc-exit_revenue_multiple-2-exit-revenue-multiple
vc-formula-vc-exit_revenue_multiple-2-exit-equity	ADD	helios-vc-exit_revenue_multiple-2-exit-equity-value	helios-vc-exit_revenue_multiple-2-exit-enterprise-value, helios-vc-exit_revenue_multiple-2-ending-cash-60
vc-formula-vc-exit_revenue_multiple-3-moic	DIVIDE	helios-vc-exit_revenue_multiple-3-gross-moic	helios-vc-exit_revenue_multiple-3-proceeds, helios-vc-exit_revenue_multiple-3-invested
vc-formula-vc-exit_revenue_multiple-3-xirr	DATED_XIRR	helios-vc-exit_revenue_multiple-3-gross-xirr	helios-vc-exit_revenue_multiple-3-cash-flow-01, helios-vc-exit_revenue_multiple-3-cash-flow-02, helios-vc-exit_revenue_multiple-3-cash-flow-03
vc-formula-vc-exit_revenue_multiple-3-ownership	DIVIDE	helios-vc-exit_revenue_multiple-3-ownership	helios-vc-exit_revenue_multiple-3-target-shares, helios-vc-exit_revenue_multiple-3-fully-diluted-shares
vc-formula-vc-exit_revenue_multiple-3-minimum-cash	MIN	helios-vc-exit_revenue_multiple-3-minimum-cash	helios-vc-exit_revenue_multiple-3-ending-cash-01, helios-vc-exit_revenue_multiple-3-ending-cash-02, helios-vc-exit_revenue_multiple-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 e9e0e615b45067af180790c3a583e025a3a4f404ff7923346e2b1580a25cf57d)
vc-formula-vc-exit_revenue_multiple-3-exit-ev	MULTIPLY	helios-vc-exit_revenue_multiple-3-exit-enterprise-value	helios-vc-exit_revenue_multiple-3-terminal-revenue, helios-vc-exit_revenue_multiple-3-exit-revenue-multiple
vc-formula-vc-exit_revenue_multiple-3-exit-equity	ADD	helios-vc-exit_revenue_multiple-3-exit-equity-value	helios-vc-exit_revenue_multiple-3-exit-enterprise-value, helios-vc-exit_revenue_multiple-3-ending-cash-60
vc-formula-vc-ordinary_cohort_nrr-1-moic	DIVIDE	helios-vc-ordinary_cohort_nrr-1-gross-moic	helios-vc-ordinary_cohort_nrr-1-proceeds, helios-vc-ordinary_cohort_nrr-1-invested
vc-formula-vc-ordinary_cohort_nrr-1-xirr	DATED_XIRR	helios-vc-ordinary_cohort_nrr-1-gross-xirr	helios-vc-ordinary_cohort_nrr-1-cash-flow-01, helios-vc-ordinary_cohort_nrr-1-cash-flow-02, helios-vc-ordinary_cohort_nrr-1-cash-flow-03
vc-formula-vc-ordinary_cohort_nrr-1-ownership	DIVIDE	helios-vc-ordinary_cohort_nrr-1-ownership	helios-vc-ordinary_cohort_nrr-1-target-shares, helios-vc-ordinary_cohort_nrr-1-fully-diluted-shares
vc-formula-vc-ordinary_cohort_nrr-1-minimum-cash	MIN	helios-vc-ordinary_cohort_nrr-1-minimum-cash	helios-vc-ordinary_cohort_nrr-1-ending-cash-01, helios-vc-ordinary_cohort_nrr-1-ending-cash-02, helios-vc-ordinary_cohort_nrr-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 c5f1b95392a0bcd773d09bfb1096b9e71847e93bb65eb14bdb463f7f4a9cb7)
vc-formula-vc-ordinary_cohort_nrr-1-exit-ev	MULTIPLY	helios-vc-ordinary_cohort_nrr-1-exit-enterprise-value	helios-vc-ordinary_cohort_nrr-1-terminal-revenue, helios-vc-ordinary_cohort_nrr-1-exit-revenue-multiple
vc-formula-vc-ordinary_cohort_nrr-1-exit-equity	ADD	helios-vc-ordinary_cohort_nrr-1-exit-equity-value	helios-vc-ordinary_cohort_nrr-1-exit-enterprise-value, helios-vc-ordinary_cohort_nrr-1-ending-cash-60
vc-formula-vc-ordinary_cohort_nrr-2-moic	DIVIDE	helios-vc-ordinary_cohort_nrr-2-gross-moic	helios-vc-ordinary_cohort_nrr-2-proceeds, helios-vc-ordinary_cohort_nrr-2-invested
vc-formula-vc-ordinary_cohort_nrr-2-xirr	DATED_XIRR	helios-vc-ordinary_cohort_nrr-2-gross-xirr	helios-vc-ordinary_cohort_nrr-2-cash-flow-01, helios-vc-ordinary_cohort_nrr-2-cash-flow-02, helios-vc-ordinary_cohort_nrr-2-cash-flow-03
vc-formula-vc-ordinary_cohort_nrr-2-ownership	DIVIDE	helios-vc-ordinary_cohort_nrr-2-ownership	helios-vc-ordinary_cohort_nrr-2-target-shares, helios-vc-ordinary_cohort_nrr-2-fully-diluted-shares
vc-formula-vc-ordinary_cohort_nrr-2-minimum-cash	MIN	helios-vc-ordinary_cohort_nrr-2-minimum-cash	helios-vc-ordinary_cohort_nrr-2-ending-cash-01, helios-vc-ordinary_cohort_nrr-2-ending-cash-02, helios-vc-ordinary_cohort_nrr-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 4441cb136d7e71709042787b1ca3677064a7bbb36bf86f6b7f2caf5c2012470b)

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-vc-ordinary_cohort_nrr-2-exit-ev	MULTIPLY	helios-vc-ordinary_cohort_nrr-2-exit-enterprise-value	helios-vc-ordinary_cohort_nrr-2-terminal-revenue, helios-vc-ordinary_cohort_nrr-2-exit-revenue-multiple
vc-formula-vc-ordinary_cohort_nrr-2-exit-equity	ADD	helios-vc-ordinary_cohort_nrr-2-exit-equity-value	helios-vc-ordinary_cohort_nrr-2-exit-enterprise-value, helios-vc-ordinary_cohort_nrr-2-ending-cash-60
vc-formula-vc-ordinary_cohort_nrr-3-moic	DIVIDE	helios-vc-ordinary_cohort_nrr-3-gross-moic	helios-vc-ordinary_cohort_nrr-3-proceeds, helios-vc-ordinary_cohort_nrr-3-invested
vc-formula-vc-ordinary_cohort_nrr-3-xirr	DATED_XIRR	helios-vc-ordinary_cohort_nrr-3-gross-xirr	helios-vc-ordinary_cohort_nrr-3-cash-flow-01, helios-vc-ordinary_cohort_nrr-3-cash-flow-02, helios-vc-ordinary_cohort_nrr-3-cash-flow-03
vc-formula-vc-ordinary_cohort_nrr-3-ownership	DIVIDE	helios-vc-ordinary_cohort_nrr-3-ownership	helios-vc-ordinary_cohort_nrr-3-target-shares, helios-vc-ordinary_cohort_nrr-3-fully-diluted-shares
vc-formula-vc-ordinary_cohort_nrr-3-minimum-cash	MIN	helios-vc-ordinary_cohort_nrr-3-minimum-cash	helios-vc-ordinary_cohort_nrr-3-ending-cash-01, helios-vc-ordinary_cohort_nrr-3-ending-cash-02, helios-vc-ordinary_cohort_nrr-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 12f1742b8a5de8fea56dbdbe287cf1bde266af06fad0c5051bcf6a05cf7403ca)
vc-formula-vc-ordinary_cohort_nrr-3-exit-ev	MULTIPLY	helios-vc-ordinary_cohort_nrr-3-exit-enterprise-value	helios-vc-ordinary_cohort_nrr-3-terminal-revenue, helios-vc-ordinary_cohort_nrr-3-exit-revenue-multiple
vc-formula-vc-ordinary_cohort_nrr-3-exit-equity	ADD	helios-vc-ordinary_cohort_nrr-3-exit-equity-value	helios-vc-ordinary_cohort_nrr-3-exit-enterprise-value, helios-vc-ordinary_cohort_nrr-3-ending-cash-60
vc-formula-vc-later_round_price-1-moic	DIVIDE	helios-vc-later_round_price-1-gross-moic	helios-vc-later_round_price-1-proceeds, helios-vc-later_round_price-1-invested
vc-formula-vc-later_round_price-1-xirr	DATED_XIRR	helios-vc-later_round_price-1-gross-xirr	helios-vc-later_round_price-1-cash-flow-01, helios-vc-later_round_price-1-cash-flow-02
vc-formula-vc-later_round_price-1-ownership	DIVIDE	helios-vc-later_round_price-1-ownership	helios-vc-later_round_price-1-target-shares, helios-vc-later_round_price-1-fully-diluted-shares
vc-formula-vc-later_round_price-1-minimum-cash	MIN	helios-vc-later_round_price-1-minimum-cash	helios-vc-later_round_price-1-ending-cash-01, helios-vc-later_round_price-1-ending-cash-02, helios-vc-later_round_price-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 94289336e5c2d10253d82ab0fdff4c2ead1c29e4398ea4b3faa37a13b98c4d99)
vc-formula-vc-later_round_price-1-exit-ev	MULTIPLY	helios-vc-later_round_price-1-exit-enterprise-value	helios-vc-later_round_price-1-terminal-revenue, helios-vc-later_round_price-1-exit-revenue-multiple
vc-formula-vc-later_round_price-1-exit-equity	ADD	helios-vc-later_round_price-1-exit-equity-value	helios-vc-later_round_price-1-exit-enterprise-value, helios-vc-later_round_price-1-ending-cash-60
vc-formula-vc-later_round_price-2-moic	DIVIDE	helios-vc-later_round_price-2-gross-moic	helios-vc-later_round_price-2-proceeds, helios-vc-later_round_price-2-invested
vc-formula-vc-later_round_price-2-xirr	DATED_XIRR	helios-vc-later_round_price-2-gross-xirr	helios-vc-later_round_price-2-cash-flow-01, helios-vc-later_round_price-2-cash-flow-02
vc-formula-vc-later_round_price-2-ownership	DIVIDE	helios-vc-later_round_price-2-ownership	helios-vc-later_round_price-2-target-shares, helios-vc-later_round_price-2-fully-diluted-shares
vc-formula-vc-later_round_price-2-minimum-cash	MIN	helios-vc-later_round_price-2-minimum-cash	helios-vc-later_round_price-2-ending-cash-01, helios-vc-later_round_price-2-ending-cash-02, helios-vc-later_round_price-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 dd3148f0eb67012e754a709f2eee08d785957ef073a88f838cladec693e76f7a)
vc-formula-vc-later_round_price-2-exit-ev	MULTIPLY	helios-vc-later_round_price-2-exit-enterprise-value	helios-vc-later_round_price-2-terminal-revenue, helios-vc-later_round_price-2-exit-revenue-multiple
vc-formula-vc-later_round_price-2-exit-equity	ADD	helios-vc-later_round_price-2-exit-equity-value	helios-vc-later_round_price-2-exit-enterprise-value, helios-vc-later_round_price-2-ending-cash-60
vc-formula-vc-later_round_price-3-moic	DIVIDE	helios-vc-later_round_price-3-gross-moic	helios-vc-later_round_price-3-proceeds, helios-vc-later_round_price-3-invested
vc-formula-vc-later_round_price-3-xirr	DATED_XIRR	helios-vc-later_round_price-3-gross-xirr	helios-vc-later_round_price-3-cash-flow-01, helios-vc-later_round_price-3-cash-flow-02
vc-formula-vc-later_round_price-3-ownership	DIVIDE	helios-vc-later_round_price-3-ownership	helios-vc-later_round_price-3-target-shares, helios-vc-later_round_price-3-fully-diluted-shares
vc-formula-vc-later_round_price-3-minimum-cash	MIN	helios-vc-later_round_price-3-minimum-cash	helios-vc-later_round_price-3-ending-cash-01, helios-vc-later_round_price-3-ending-cash-02, helios-vc-later_round_price-3-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 3e80bb8504675fbc16fe12c3decb805ba21fb73cffe36d001b9ec95e273d1396)
vc-formula-vc-later_round_price-3-exit-ev	MULTIPLY	helios-vc-later_round_price-3-exit-enterprise-value	helios-vc-later_round_price-3-terminal-revenue, helios-vc-later_round_price-3-exit-revenue-multiple
vc-formula-vc-later_round_price-3-exit-equity	ADD	helios-vc-later_round_price-3-exit-equity-value	helios-vc-later_round_price-3-exit-enterprise-value, helios-vc-later_round_price-3-ending-cash-60
vc-formula-vc-milestone_state-1-moic	DIVIDE	helios-vc-milestone_state-1-gross-moic	helios-vc-milestone_state-1-proceeds, helios-vc-milestone_state-1-invested
vc-formula-vc-milestone_state-1-xirr	DATED_XIRR	helios-vc-milestone_state-1-gross-xirr	helios-vc-milestone_state-1-cash-flow-01, helios-vc-milestone_state-1-cash-flow-02

FORMULA	OPERATION	OUTPUT METRIC	OPERANDS
vc-formula-vc-milestone_state-1-ownership	DIVIDE	helios-vc-milestone_state-1-ownership	helios-vc-milestone_state-1-target-shares, helios-vc-milestone_state-1-fully-diluted-shares
vc-formula-vc-milestone_state-1-minimum-cash	MIN	helios-vc-milestone_state-1-minimum-cash	helios-vc-milestone_state-1-ending-cash-01, helios-vc-milestone_state-1-ending-cash-02, helios-vc-milestone_state-1-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 bf9cfa93b6e6c926b2fe6c0cf9c4b1a4675627551420ae5a9afc483db06acfea)
vc-formula-vc-milestone_state-1-exit-ev	MULTIPLY	helios-vc-milestone_state-1-exit-enterprise-value	helios-vc-milestone_state-1-terminal-revenue, helios-vc-milestone_state-1-exit-revenue-multiplier
vc-formula-vc-milestone_state-1-exit-equity	ADD	helios-vc-milestone_state-1-exit-equity-value	helios-vc-milestone_state-1-exit-enterprise-value, helios-vc-milestone_state-1-ending-cash-60
vc-formula-vc-milestone_state-2-moic	DIVIDE	helios-vc-milestone_state-2-gross-moic	helios-vc-milestone_state-2-proceeds, helios-vc-milestone_state-2-invested
vc-formula-vc-milestone_state-2-xirr	DATED_XIRR	helios-vc-milestone_state-2-gross-xirr	helios-vc-milestone_state-2-cash-flow-01, helios-vc-milestone_state-2-cash-flow-02, helios-vc-milestone_state-2-cash-flow-03
vc-formula-vc-milestone_state-2-ownership	DIVIDE	helios-vc-milestone_state-2-ownership	helios-vc-milestone_state-2-target-shares, helios-vc-milestone_state-2-fully-diluted-shares
vc-formula-vc-milestone_state-2-minimum-cash	MIN	helios-vc-milestone_state-2-minimum-cash	helios-vc-milestone_state-2-ending-cash-01, helios-vc-milestone_state-2-ending-cash-02, helios-vc-milestone_state-2-ending-cash-03; +57 more in model-appendix.json (ordered-list SHA-256 0b0946538a5afbfe72d79ab464226fb755655557adaef8bbfd0e4185969190a)
vc-formula-vc-milestone_state-2-exit-ev	MULTIPLY	helios-vc-milestone_state-2-exit-enterprise-value	helios-vc-milestone_state-2-terminal-revenue, helios-vc-milestone_state-2-exit-revenue-multiplier
vc-formula-vc-milestone_state-2-exit-equity	ADD	helios-vc-milestone_state-2-exit-equity-value	helios-vc-milestone_state-2-exit-enterprise-value, helios-vc-milestone_state-2-ending-cash-60
vc-formula-value-ordinary-expansion-minimum_cash_delta_cents	SUBTRACT	helios-value-ordinary-expansion-minimum_cash_delta_cents	helios-value-ordinary-expansion-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-ordinary-expansion-target_proceeds_delta_cents	SUBTRACT	helios-value-ordinary-expansion-target_proceeds_delta_cents	helios-value-ordinary-expansion-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-ordinary-expansion-gross-xirr-delta	SUBTRACT	helios-value-ordinary-expansion-gross-xirr-delta	helios-value-ordinary-expansion-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-ordinary-expansion-gross-moic-delta	SUBTRACT	helios-value-ordinary-expansion-gross-moic-delta	helios-value-ordinary-expansion-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-optimizer-unit-economics-minimum_cash_delta_cents	SUBTRACT	helios-value-optimizer-unit-economics-minimum_cash_delta_cents	helios-value-optimizer-unit-economics-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-optimizer-unit-economics-target_proceeds_delta_cents	SUBTRACT	helios-value-optimizer-unit-economics-target_proceeds_delta_cents	helios-value-optimizer-unit-economics-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-optimizer-unit-economics-gross-xirr-delta	SUBTRACT	helios-value-optimizer-unit-economics-gross-xirr-delta	helios-value-optimizer-unit-economics-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-optimizer-unit-economics-gross-moic-delta	SUBTRACT	helios-value-optimizer-unit-economics-gross-moic-delta	helios-value-optimizer-unit-economics-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-sales-governance-minimum_cash_delta_cents	SUBTRACT	helios-value-sales-governance-minimum_cash_delta_cents	helios-value-sales-governance-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-sales-governance-target_proceeds_delta_cents	SUBTRACT	helios-value-sales-governance-target_proceeds_delta_cents	helios-value-sales-governance-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-sales-governance-gross-xirr-delta	SUBTRACT	helios-value-sales-governance-gross-xirr-delta	helios-value-sales-governance-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-sales-governance-gross-moic-delta	SUBTRACT	helios-value-sales-governance-gross-moic-delta	helios-value-sales-governance-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-combined-minimum_cash_delta_cents	SUBTRACT	helios-value-combined_minimum_cash_delta_cents	helios-value-combined-minimum_cash_delta_cents-result, helios-value-base-minimum_cash_delta_cents
vc-formula-value-combined-target_proceeds_delta_cents	SUBTRACT	helios-value-combined_target_proceeds_delta_cents	helios-value-combined-target_proceeds_delta_cents-result, helios-value-base-target_proceeds_delta_cents
vc-formula-value-combined-gross-xirr-delta	SUBTRACT	helios-value-combined-gross-xirr-delta	helios-value-combined-gross-xirr-delta-result, helios-value-base-gross-xirr-delta
vc-formula-value-combined-gross-moic-delta	SUBTRACT	helios-value-combined-gross-moic-delta	helios-value-combined-gross-moic-delta-result, helios-value-base-gross-moic-delta
vc-formula-value-standalone-proceeds-sum	SUM	helios-value-sum_standalone_target_proceeds_delta_cents	helios-value-ordinary-expansion-target_proceeds_delta_cents, helios-value-optimizer-unit-economics-target_proceeds_delta_cents, helios-value-sales-governance-target_proceeds_delta_cents
vc-formula-value-interaction	SUBTRACT	helios-value-interaction_residual_cents	helios-value-combined_target_proceeds_delta_cents, helios-value-sum_standalone_target_proceeds_delta_cents

Reproducibility boundary

All data and results are synthetic. Exact arithmetic, source locators, scenario receipts, and deterministic generation establish internal reproducibility only. They do not establish live-deal accuracy, investment approval, or real-world performance.